

Question number	Answer	Additional guidance	Mark
7(a)(i)	An answer electron (microscope) (1)	REJECT electric, electrical, E, electronic ACCEPT EM or EM REJECT SEM \ scanning electron microscope	(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	A calculation showing the following steps: - calculation of length in micrometres of A-B (1) - answer given to three significant figures (1)	No ecf 30 000 μm Accept 29000-31000 $(30\,000 \div 23\,500) = 1.28$ $= 1.28\,(\mu\text{m})$ Accept 1.23 – 1.32 μm Correct answer gains full marks	(2)

Question number	Answer	Additional guidance	Mark
7(b)(i)	B is the correct answer A is not the correct answer as chemiosmosis is not where hydrogen ions are pumped from the matrix to the intermembrane space through ATP synthase C is not the correct answer as chemiosmosis is not where hydrogen ions diffuse from the intermembrane space to the stroma through ATP synthase D is not the correct answer as chemiosmosis is not where hydrogen ions jump from the inner membrane to the outer membrane and out through ATP synthase		(1)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	B is the correct answer A is not the correct answer as Krebs cycle activity decreases C is not the correct answer as Krebs cycle activity and oxygen consumption decrease D is not the correct answer as oxygen consumption decreases		(1)

Question number	Answer	Additional guidance	Mark
7(b)(iii)	A is the correct answer B is not the correct answer as oxidised NADP is not used in the removal of a C atom from pyruvate in the link reaction C is is not the correct answer as reduced NAD is not used in the removal of a C atom from pyruvate in the link reaction D is not the correct answer as reduced NADP is not used in the removal of a C atom from pyruvate in the link reaction		(1)

Question number	Answer	Additional guidance	Mark
7(c)	• carbon dioxide / CO₂ (1) • ATP (1) • reduced NAD / NADH (1) • reduced FAD / FADH₍₂₎ (1) • oxaloacetate / OAA (1)	mark first 3 Any three equals two marks two equals one mark IGNORE incorrect formula IGNORE numbers e.g. 6ATP	(2)

Question number	Answer	Additional guidance	Mark
7(d)	A description that includes four of the following points: - as the {number of days / times} increases, the number of mitochondria increases (1) - as the {number of days / times} increases, the concentration of ATP increases (1) - due to increased demand for ATP in {cell activity / cell processes} (1) - ATP {production / use} does not increase in the same way as the number of mitochondria (1) - comment on {error bars / standard deviation / lack of data after 10 divisions / data missing} (1)	ACCEPT positive correlation between time and number of mitochondria Must be after cell division NOT as the cell divides ACCEPT positive correlation e.g. other described processes in {cell differentiation / neurones} that requires energy from ATP e.g. protein synthesis / production of sodium ion channels / neurotransmitters / receptors / elongation of cell / establishing resting potential ACCEPT smaller (%) increase for ATP increase / converse e.g. error bars for number of mitochondria do not overlap so results are significantly different Error bars for concentration of ATP overlap {1-3 days / 5-7 days / 7-10 days} so results not significantly different ACCEPT converse for significant difference - eg mitochondria 7-10 days IGNORE sample size and other comments on methodology	(4)

